

Chapter 18 – Interpolation 插值 Cheatsheet

◆ 18.1 Linear Interpolation 线性插值

✦ 公式:

$$f_1(x) = f(x_0) + \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] \cdot (x - x_0) \quad \text{或} \quad f_1(x) = b_0 + b_1(x - x_0)$$

✦ 真值误差:

$$\varepsilon_t = \left| \frac{f(x) - f_1(x)}{f(x)} \right| \times 100\%$$

◆ Q18-1 示例:

- 数据: $x_0 = 8, f(x_0) = \log 8 = 0.9030900$
- $x_1 = 12, f(x_1) = \log 12 = 1.0791812$
- 求 $f(10) \Rightarrow f_1(10) = 0.991136$
- 真值 $\log 10 = 1 \Rightarrow \varepsilon_t = 0.886\%$

◆ 18.2 Newton's Divided-Difference Interpolating Polynomials 牛顿插值法

✦ 二次插值 (二阶) 公式:

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

- $b_0 = f(x_0)$
- $b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$
- $b_2 = \frac{\left(\frac{f(x_2) - f(x_1)}{x_2 - x_1} - b_1 \right)}{x_2 - x_0}$

◆ Q18-2 示例:

- $x = 10$, 使用点: $x_0 = 9, x_1 = 11, x_2 = 8$
- $f_2(10) = 1.0003434$, 真值: 1
- $\varepsilon_t = 0.034\%$

✦ 三次插值 (三阶) 公式:

$$f_3(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$

差商 (Divided Differences) 递推关系:

- 一阶:
 $f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$
- 二阶:
 $f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$
- 三阶:
 $f[x_3, x_2, x_1, x_0] = \frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0}$

✦ 误差估计公式:

$$R_n \approx f[x_{n+1}, x_n, \dots, x_0] \cdot (x - x_0)(x - x_1) \dots (x - x_n)$$

◆ 18.3 Lagrange Interpolation 拉格朗日插值

✦ 一阶:

$$f_1(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1)$$

✦ 二阶:

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

✦ 三阶:

$$f_3(x) = \sum_{j=0}^3 \left[\prod_{\substack{i=0 \\ i \neq j}}^3 \frac{x - x_i}{x_j - x_i} \right] f(x_j)$$

✦ 注意: 每增加一个点, 公式需完全重写!

◆ 18.4 Inverse Interpolation 逆插值法

◆ Q18-11 示例:

- 数据: $x = 3, 4, 5, 6, f(x) = 0.3333, 0.25, 0.2, 0.1667$
- 使用四个点构建三次插值:

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

- 解得:

$$f(x) = 0.949 - 0.329883x + 0.04985x^2 - 0.002767x^3$$

- 求解 $f(x) = 0.23$ 用二分法: 结果为 $x = 4.34130$

◆ 18.5 Newton-Gregory Forward Interpolation 牛顿-格雷戈里前向差分插值

✦ 特点:

- 适用于等间距数据
- 使用差分表 (forward difference table)

◆ 18.6 Spline Interpolation 样条插值

◆ 18.6.1 Linear Splines 一阶样条

📌 公式:

在线性段 $[x_i, x_{i+1}]$ 上:

$$f(x) = f(x_i) + \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}(x - x_i)$$

◆ E18-8 示例:

- 数据: $x = 4.5, 7.0, f(x) = 1.0, 2.5$
- $x = 5$ 所属区间为 $[4.5, 7.0]$
- 斜率:

$$m = \frac{2.5 - 1.0}{7.0 - 4.5} = 0.60$$

- 插值值:

$$f(5) = 1.0 + 0.6 \cdot (5.0 - 4.5) = 1.3$$